



OPEN Multi-modal emotional analysis in customer relation management and enhancing communication through integrated affective computing

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An important part of customer relationship management (CRM) is being able to read emails for emotional cues; this helps with both communication and keeping customers satisfied. This study aims to improve email emotion identification by creating a system combining visual clues, aural signals, and textual information. To analyze text and emoji, the system uses advanced affective computing techniques such as Robustly Optimized Bidirectional Encoder Representations from Transformers Approach (RoBERTa), Convolutional Neural Networks (CNN) for images, Bidirectional Convolutional Long Short-Term Memory (BiConvLSTM) for video, and Cross-Modal BERT for audio. Together, they enable a wider variety of emotional signals to be extracted and understood from email content, yielding more insightful results than text-based analysis could on its own. By facilitating better two-way communication and customer satisfaction, the technology intends to improve CRM by providing actionable information that firms can use to personalize responses. This research delves into the possible uses of multi-modal emotional analysis across different businesses dealing with customers and builds a strong foundation.

Keywords Multi-modal emotional analysis, Affective computing, Visual cues, Auditory signals, Textual information, CRM

Multi-modal emotional analysis is a relatively new yet critically important branch of emotional psychology¹. A multi-modal framework simultaneously supports several input and output channels, allowing for an interactive paradigm reminiscent of human speech². With additional options for modalities, users may have more natural conversations, which boosts their immersion and the efficacy of their communications³. Studies on auditory emotional cognition, facial affective interpretation, and sentiment discernment in textual information have amassed substantial scientific literature⁴. Multiple views of emotions (emotions as expressions, emotions as embodiments) agree that feeling an emotion triggers a cascade of reactions in our bodies and minds⁵. To rephrase, there are several channels via which an individual's emotions might be conveyed; therefore, it is only reasonable to read signals from all of these channels concurrently to enhance emotion identification⁶. Focusing on just one of these modalities might lead us astray when trying to understand a person's emotional state. The physiological signals of an individual may be low even though they are smiling⁷.

A person's voice may be very nervous even though they claim to be comfortable, and someone can be dead serious yet on the verge of a panic attack and write about being incredibly excited about something while being emotionless⁸. A common thread runs across all of these instances and risks missing important details if focused on just one-way people show their emotions⁹. Spectrograms reveal unique energy patterns representing emotions, especially in audible forms like voice¹⁰. Spectrogram analysis reveals these patterns since they exhibit distinct energy distributions across various frequencies¹¹. CNN easily highlights Data points with high activity or abrupt changes¹². This feature is useful for emotion analysis because CNN is so good at capturing fast-paced,

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intense emotions¹³. In scenarios where emotions could spike rapidly or peak momentarily, this targeted analysis allows the model to distinguish between weak and strong emotional signals, guaranteeing a complete picture of the emotional terrain¹⁴.

Another important function of text is as a medium for conveying and understanding feelings¹⁵. In written language, word choice, phrasing, and punctuation provide subtle indications that allow writers to communicate a broad range of emotions, from happiness and enthusiasm to sorrow and despair¹⁶, in contrast to spoken words and bodily reactions, which typically provide clear emotional clues. However, gathering large amounts of accurate data is a huge obstacle to building text-based models¹⁷. Large, task-specific datasets must be readily available for these models to function accurately and efficiently¹⁸. Developers and researchers may have functional emotion detectors but lack the resources to create a system that collects them all¹⁹. To address the dearth of practical framework implementations in the existing literature, BiConvLSTMis introduced a framework that will help to bridge a gap in customer relationship management²⁰. BiConvLSTM is dedicated to extracting spatial and temporal aspects from video inputs, Cross-Modal BERT examines audio signals for linguistic and affective patterns, and RoBERTa evaluates text-based data, such as conversation transcripts, presumably processing contextual and semantic emotional cues. Understanding might be improved with a unified flow that processes inputs (video, audio, and text) via their respective models. Then, a fusion layer could integrate all three modalities into a holistic emotional representation.

The main objective of this paper is as follows:

- **Develop a comprehensive multi-modal emotion detection system:**

Use state-of-the-art methods such as RoBERTa for text and emoji analysis, CNN for image processing, BiConvLSTM for video analysis, and Cross-Modal BERT for audio to build a sophisticated emotional analysis system that combines visual, aural, and textual input. Emotional tones in email conversations may be better understood using this technique.

- **Enhance customer relationship management (CRM) through personalization:**

- The multi-modal emotional analysis method provides useful insights that help firms tailor their communication strategy to each individual. The solution will enhance two-way communication by comprehending the emotional context of consumer encounters, resulting in higher customer satisfaction.

- **Explore Industry applications and impact:**

Develop a solid plan for implementing multi-modal emotional analysis into customer relationship management systems in sectors that interact with customers. To enhance customer service and happiness across various company contexts, this research will investigate the possible advantages, disadvantages, and consequences of adopting such a system.

This article is organized as follows: Sect. 2 considers the related study of multi-modal emotional analysis. In Sect. 3, the suggested model of BiConvLSTMis is explained. In Sect. 4, the efficiency of BiConvLSTMis is discussed, and finally, Sect. 5 concludes the paper with a future study.

Related work

Emotion and sentiment analysis have emerged as a new social media trend, allowing users to comprehend product opinions expressed. As social media continues to grow in popularity and technology improves, more and more product opinions are being expressed in video format. People are recording themselves using webcams or other devices to share their thoughts on platforms like YouTube or Facebook to inform others about these products before they buy. Video reviews often include pros and cons lists, comparisons to competing brands, and other such information.

Abdullah, S. M. S. A. et al.²¹, to provide a smooth user experience, recent studies in human-computer interaction have focused on considering the customer's emotional state. As a result, humans have a fighting chance and can be useful in many areas, such as healthcare and education. Expressions, facial pictures, physiological signals, and neuroimaging approaches explain how human emotions may define various methods. This study summarizes and compares the current research on emotional identification of multi-modal data using DLT.

The exponential expansion of multi-modal content in the modern, competitive corporate world produces massive unstructured data. Text, music, pictures, and video are all examples of large unstructured data that do not adhere to any certain format or organization. The difficulties of sentiment and emotion modeling result from this article's multi-modal, unstructured big data. Seng, J. K. P. et al.²² provide an overview of emotion and sentiment modeling that covers ground-breaking methods. This paper incorporates many modalities to provide a fresh framework for big data sentiment and emotion modeling.

Kaur, S., & Sharma, R. et al.²⁰ proposed a new way of thinking about how people engage with technology, which has emerged due to recent developments in digital transformation and artificial intelligence (AI). Given the widespread deployment of AI in many business operations, organizations are confronted with the tremendous problem of integrating AI with emotional intelligence (EI). More and more technology is engaged, becoming a two-way street; that much is certain. Both humans and machines learn to interact with one another via training.

Sentiment analysis is gaining insight into and making sense of people's feelings, thoughts, and perspectives on a specific topic. Due to the proliferation of e-commerce sites, people can express their ideas freely and in real time on social media. To handle such subjective input, fuzzy logic is necessary since the real-world data is imprecise and nebulous. Due to the subjective character of views and their meanings might vary, fuzzy logic has proven to be a useful tool for capturing these expressions by Vashishtha, S. et al.²³.

Social media platforms like Facebook and YouTube are seeing a dramatic increase in the shared volume of multi-modal content. The rapid expansion of multi-modal data necessitates immediately creating a smart multi-modal analysis framework capable of efficiently extracting information from many modalities. In applications including e-learning, e-health, automated video content tagging, and human-computer interaction, Poria S. et al.²⁴ present a new multi-modal information extraction agent that can infer and collect the emotional and semantic information linked with user-generated multi-modal data.

Fei Wang et al.²⁵ suggested the Geometric algebra-enhanced network for skin lesion detection with diagnostic prior. The proposed system is divided into four separate modules: skin area focus, skin feature extraction, anomalous feature attention, and skin diagnostic interpretation. It combines medical diagnostic priors and aims to increase clinical credibility. These units directly reflect the actual procedures used in clinical diagnostics. The development of neural networks and an attention module based on geometric algebra allowed us to get more structural information and richer characteristics to aid in categorizing skin lesions. Reduced to 25% of their original size, the parameters of the geometric algebra-based network and the attention mechanism make the skin lesion detection network computationally light and efficient. When tested on the publicly available ISIC2020 skin picture dataset, the solution beats the state-of-the-art. Over the baseline, the technique improves accuracy by 2.5% and area under the curve by 1.2%, reaching 98.8%.

Xianxun Zhu et al.²⁶ recommended the Non-contact Single/Multiple Emotional and Behavioral State Assessment Methods based on Client-Server Based Recognition System. This work presents a multi-modal emotion and behavior identification system based on client-server architecture. This approach combines many modalities, such as speech, text, facial expressions, and movement, for a complete evaluation of mental states without touching the subject. It greatly improves the speed and quality of evaluations by customizing analysis to diverse circumstances' demands and generating personalized diagnostic results. The author has also created a user-friendly visual interface to improve the effectiveness of diagnostics and the precision of decisions. In real-world hospital settings, the author verified and tested our technology, which satisfies the basic needs of doctors for mental state analysis.

Rui Wang et al.²⁷ proposed the Multi-modal emotion recognition using tensor decomposition fusion and self-supervised multi-tasking. The approach begins by reducing the model's parameter count using Tucker decomposition methods, which reduces the danger of overfitting. Consequently, it better captures the emotional distinctions across modalities by constructing a learning mechanism for both unimodal and multi-modal tasks and including the idea of label creation. Results from our comprehensive testing and analysis on publicly available datasets such as CMU-MOSI and CMU-MOSEI demonstrate that the technique substantially outperforms state-of-the-art methods.

Xianxun Zhu et al.²⁸ discussed the Key Technologies for Emotion Analysis Using Multi-modal Information. Analysis of preprocessing, feature extraction, and tools applicable to voice, text, pictures, and physiological inputs is the main focus of unimodal processing approaches. Early, late, model, and hybrid fusion tactics are highlighted as the author delves further into multi-modal data fusion's intricacies. Key discoveries show the importance of understanding emotions across different modalities. Issues of representation models, expressiveness, and emotion elicitation are covered in depth. In addition, the author finds difficulties in dealing with missing modalities, restricted data situations, creating datasets, synchronizing modalities, and optimizing models. To tackle these difficulties, the author provides practical answers and recommendations. Multi-modal emotion analysis has many potential uses, from medical assessments to driver mood detection.

Xianxun Zhu et al.²⁹ deliberated on Emotional recognition based on brain-like multi-modal hierarchical perception. To improve performance, this framework combines cognitive processes with algorithms for machine learning. First, the author models our model based on how the human brain processes emotional signals by extracting deep features. These characteristics are combined via a cross-attention method to investigate potential connections across different modes. Finally, the gathered emotional data is labeled and identified. Our method demonstrates its efficacy and provides a new angle on multi-modal emotion detection since experimental findings reveal an average recognition accuracy of 82% across four separate emotion categories.

Tahani Alsaedi et al.³⁰ suggested the Transformer-based Deep Learning Model for Sentiment Mining in E-Commerce. The suggested framework uses these cutting-edge methods to classify user input as good, negative, or neutral. To build a sentiment analysis system that is both powerful and reliable, yielding valuable insights that may propel organizational growth and boost consumer happiness. This framework's main processes include gathering data, extracting features using natural language processing enhancements, detecting sentiment, and finally, classifying the data using finite-state automata. The author evaluated our NLP-centric method on several e-commerce sector customer feedback datasets to see how well it worked. The system's performance was evaluated using recall, precision, and accuracy criteria. The findings show that the suggested framework is successful, with an accuracy rate of 93.75% and an improvement over current benchmark approaches.

Taylor Jing Wen et al.³¹ proposed Infusing Affective Computing Models into Advertising Research on Emotions. This article explores potential ways to include affective computational methodologies in emotion-based advertising research. Here, the author examines how emotional computing works across three distinct modalities: text, images, and audio. The author then showcases several advertising research instances using each modality and some computational tools. To help researchers in the field of advertising assess and choose the best affective computing models, the author provides an evaluation methodology and reviews many state-of-the-art multi-modal tools for emotion identification. This study wraps up by discussing how affective computing techniques might help further theory and research into emotion-based advertising by filling in some of the gaps that have been identified. The growing area of computational advertising research relies on this article's theoretical understanding, ethical issues, and practical recommendations to enhance its methodology.

In summary, advanced methods in machine learning, deep learning, artificial intelligence, fuzzy neural networks, and big data analysis are covered in the books that are offered. These techniques are finding more and

more uses in fields as diverse as healthcare, education, and business, where they aim to improve human-computer interaction by considering sentimental and emotional factors. Emotion detection and sentiment analysis may greatly benefit from incorporating multi-modal data, including text, visuals, and physiological signs, as shown in the works. In these situations, unstructured, large-scale data presents unique obstacles. New frameworks and algorithms are being developed to tackle these issues, such as agents for multi-modal information extraction and techniques based on fuzzy logic.

Proposed method

This study has seen remarkable progress in emotion analysis, a crucial component of human-machine interactions. The increasing prevalence of multi-modal data sources, including text, pictures, and audio, highlights the critical need to thoroughly examine key components in this field. Our research explores the world of emotion analysis in great detail by examining multi-modal data sources that include voice, text, visuals, and physiological signals. It provides a hand-picked selection of the most relevant articles, online communities, and contests. Analysis of preprocessing, feature extraction, and tools applicable to voice, text, pictures, and physiological inputs is the main focus of unimodal processing approaches.

Contribution 1: develop a comprehensive multi-modal emotion detection system

Extraction, expression, and representation models of emotions are covered in depth. In addition, it finds difficulties in dealing with missing modalities, restricted data situations, creating datasets, synchronizing modalities, and optimizing models. To overcome these obstacles, realistic recommendations and solutions are offered. Many fields rely on multi-modal emotion analysis; for example, medical assessments and driver sentiment detection are only two examples. Both academics and businesspeople may benefit from our extensive review. It elucidates where things are regarding research and points toward possible avenues for future advances.

Figure 1 shows an application that can detect emotions using multiple senses. Audio, facial expressions, and physiological signs are the data collected first. Preprocessing involves reducing noise, detecting faces, and extracting features from the acquired data. A model is trained using the processed data after preprocessing; the model is then verified and fine-tuned. Multi-modal fusion is the next step in developing an emotion recognition model; it entails merging the various data streams. This model is used for emotion classification to determine the subject's unique emotional state. Some of the many uses for the evaluated categorization findings include emotion detection, measuring customer happiness, and refining communication techniques. The end aim is a strong multi-modal emotion identification system that can improve human-computer interactions and better comprehend human emotions. This system can handle a wide variety of real-world situations.

$$M_{Kp} = V(0) + N(T-1) - f^{t-yp} * Rm_{np} - (T - (kp)) \quad (1)$$

Equation 1, $V(0)$ depicts the starting values of emotions, $N(T-1)$ shows emotional drift function capture change over $(T-1)$ time steps, f^{t-yp} is the function that makes temporal adaptations to emotions, and Rm_{np} multi-modal interaction effects, M_{Kp} denotes the composite emotion vector for modality. It is possible to account for temporal aspects in detecting emotions using the expression $T - (kp)$.

$$S^{rt} = D_f(b - np) + Max(b_2, Up) + Fb^{(u-1)} \quad (2)$$

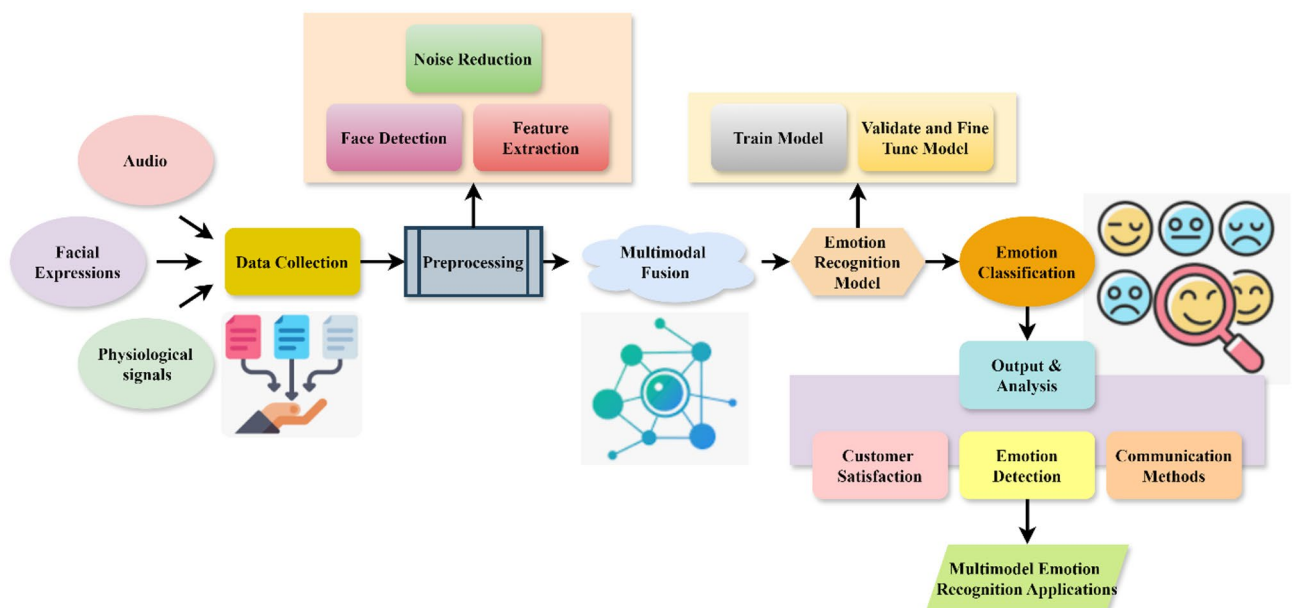


Fig. 1. Multi-modal emotion recognition application.

Here, the Eq. 2 that reflects S^{rt} the differential influence of mood indicators is $D_f(b - np)$. The greatest feeling signals across both modalities are captured by $Max(b_2, Up)$, and loop changes based on prior research are accounted for by $Fb^{(u-1)}$.

$$F_{g(1-p)} = D(v(N - er)) * G(t) - U(p - qw) \quad (3)$$

The accounted affective signal calculated from changing inputs $F_{g(1-p)}$ is represented by Eq. 3, $D(v(N - er))$ are the time-dependent elements impacting the emotion analyses are denoted by temporal gain $G(t)$, and inconsistencies or noise in the data are compensated for by $U(p - qw)$.

$$b(y, z) = f^{-kp} * v^{(m-n)} + m_{vp}(j - m) \quad (4)$$

In this context, the Eq. 4, f^{-kp} represents the inverse impact of the forecasting model on mental assessment, $m_{vp}(j - m)$ represents the variation across approaches, and $b(y, z)$ represents modifications derived from feedback unique $v^{(m-n)}$ to any given cross-modality difference.

Figure 2 shows a neural network-based method for emotion identification in action. Two networks, the Xception, and the BiConvLSTM networks, analyze input data, including face photographs, from the outset. The purpose of these networks is to obtain useful information from face data. The extracted features are subjected to an intersection operation to determine whether the intersection unit is active. After that, sort the intersection findings by mood, labeling them good or negative. Feelings of joy and contentment are examples of positive emotions, whilst feelings of despair and rage are examples of negative emotions. The outcome of this procedure is to classify the detected emotion into four categories: happy, neutral, sad, or angry. The robot picture suggests that this technology might be a component of a larger framework for emotion detection, which could be incorporated into a robotic system. To have more organic and emotionally intelligent interactions with people, the robot utilizes the labeled emotions to modify its reactions or actions.

$$\frac{E}{f(m^2)k - t} = [\forall_2(U - Pk)] - \left(1 + \left(\frac{U(q - w)}{2}\right)\right) \quad (5)$$

Here, the normalized emotional reaction is represented by the Eq. 5, $\forall_2(U - Pk)$, while the aggregate influence of many emotional modalities is captured by $\frac{E}{f(m^2)k - t}$. Additional parameters that impact emotion detection are taken into account by the $1 + \frac{U(q - w)}{2}$.

$$R_{vt} = Cv(m - nT^2) * (\partial_2(L - Kpm)) \quad (6)$$

The Eq. 6, R_{vt} ties into the proposed multi-modal emotional analysis system by quantifying the response accuracy based on different emotional inputs. Here, $Cv(m - nT^2)$ represents a weighted factor that adjusts for emotional variance over time while $\partial_2(L - Kpm)$ denotes the sensitivity of the system to changes in emotional signals.

$$\frac{Ef}{G(m - n)} = V(e - Fp) * Mn^{p-k}(r - Z^{pm}) \quad (7)$$

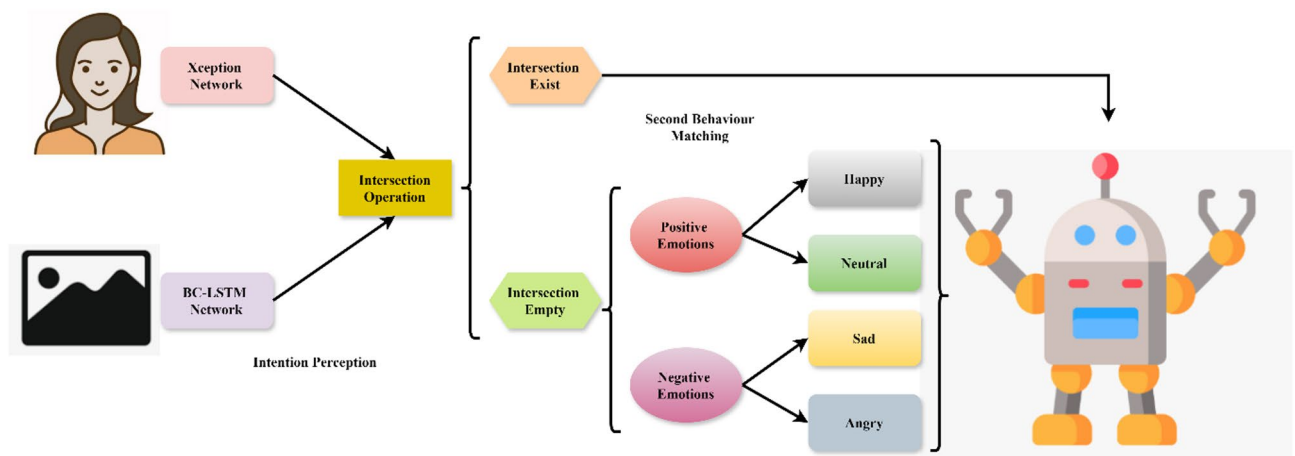


Fig. 2. System flow diagram for multi-modal emotion recognition.

Here, the emotional analysis's efficacy is represented by the Eq. 7, $\frac{Ef}{G(m-n)}$, while the variation in emotional inputs is captured by $V(e - Fp)$. The effect of modality-specific modifications $(r - Z^{pm})$ and time-related variables are shown by the expression $\setminus Mn^{p-k}$.

An individual feature vector is generated by each modality and given a weight that changes depending on the signal intensity and contextual significance at every particular moment. An attention-based gating mechanism is used to calculate these weights, giving greater emotional confidence ratings to modalities. A weighted summation of normalized vectors is used to perform the fusion process, which enables the model to minimize noise or ambiguity from weaker modalities and accentuate more trustworthy emotional signals. System performance is greatly affected by changes to these weights; for example, if the system relies too much on low-signal modalities, such as blurry video or loud audio, it might lead to misclassification, while a balanced weighting approach improves accuracy and resilience. Emotion recognition accuracy in practical CRM settings is enhanced by this adaptive weighting technique, which keeps the system responsive to changing input quality across modalities and context-sensitive.

Contribution 2: enhance customer relationship management through personalization

Pattern recognition and human-computer interaction rely heavily on facial expressions because of the wealth of emotional data they provide. Emotion detection is slowly moving into the spotlight of voice recognition technologies. This study suggests a fusion model for picture and audio recognition to improve the accuracy of emotion detection, as focusing on single-modal analysis does not provide particularly accurate results.

The method of extracting emotional information from multimedia email inputs (including text, photos, video, and audio) is shown in Fig. 3. The procedure starts with the input step, which involves receiving an email with several forms of material. The input is next processed at the preprocessing step, which involves preparing and normalizing the text, photos, video, and audio for further analysis. Data goes into modal analysis after preprocessing. This study leverages various models, such as RoBERTa for text, CNN for pictures, bi-LSTM for sequential data, and Cross-Modal BERT for integrating information across multiple modalities.

$$S_0 = C^{v1} + Mt_{p-1} * (b - Cp) - V_2M(n - Wr) \quad (8)$$

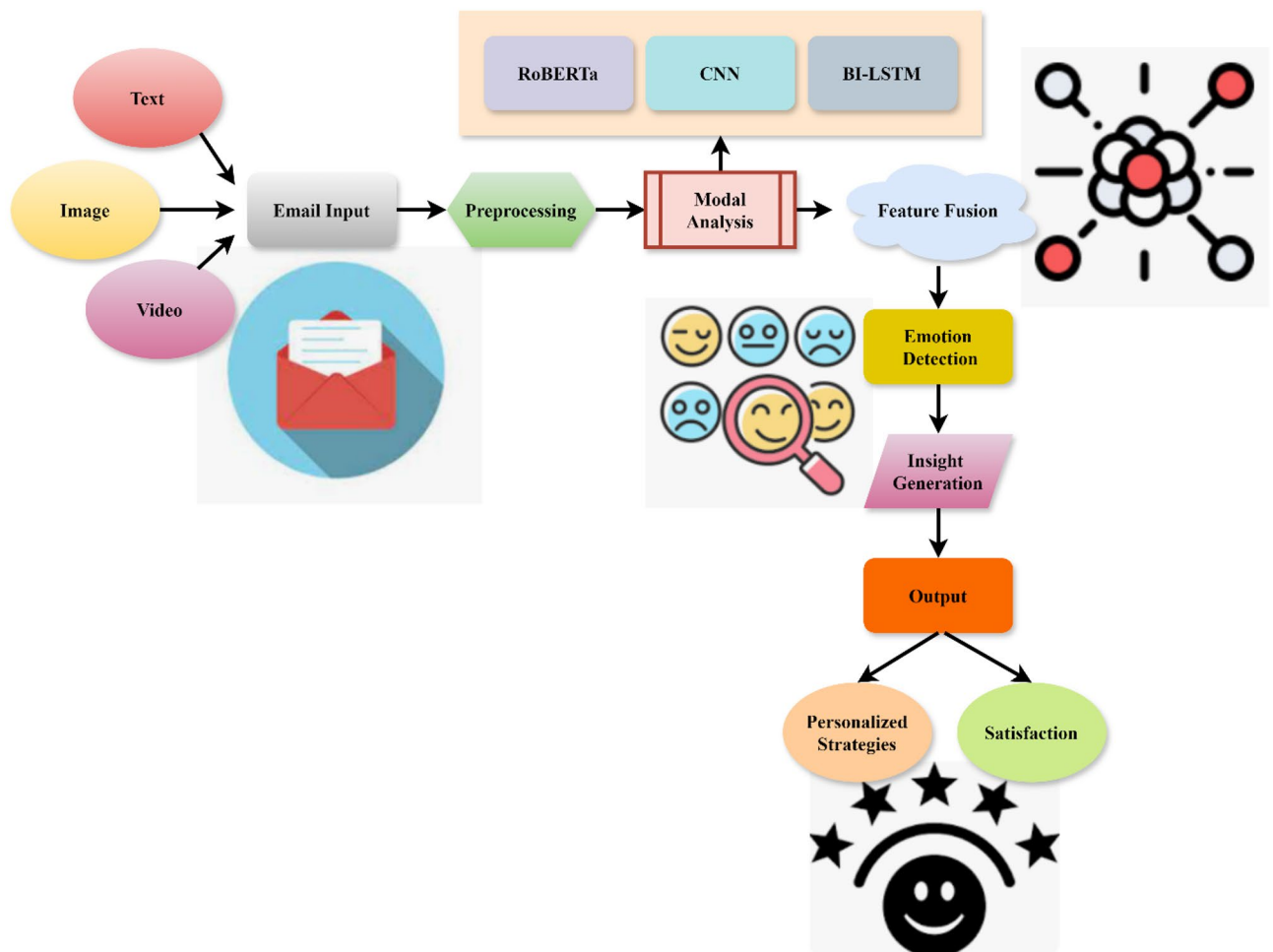


Fig. 3. Emotion detection and insight generation from multi-modal email inputs.

In this case, the Eq. 8, C^{v1} denotes a starting constant for the detection of emotions ($b - Cp$), while the equation Mt_{p-1} takes into consideration modifications S_0 depending on past emotional outputs and returns. The term $V_2M(n - Wr)$ accounts for additional modifications related to modality (e.g., input sources like voice or text) and temporal effects (time-sensitive variations in emotions). Together, these components contribute to a dynamic score S_0 , which evolves based on both historical and real-time inputs.

Customer relationship management is an old philosophy that has existed since businesses began. CRM is an approach to business that aims to strengthen bonds with consumers to meet their demands and boost revenue, customer value, and service quality. Although customer relationship management is not a new idea, it has grown in importance due to the ever-increasing need for personalized products and services. According to Chen and Popovich, customer relationship management connects front- and back-office operations with customers. CRM integrates many customer-centric management strategies to establish a lasting connection with customers. A customer relationship management system is useful for various corporate functions, including data warehousing, sales force automation, data mining, reporting tools, decision assistance, and data support. Operational, analytical, and collaborative CRM systems are the three basic categories, as seen in Fig. 4. The operational

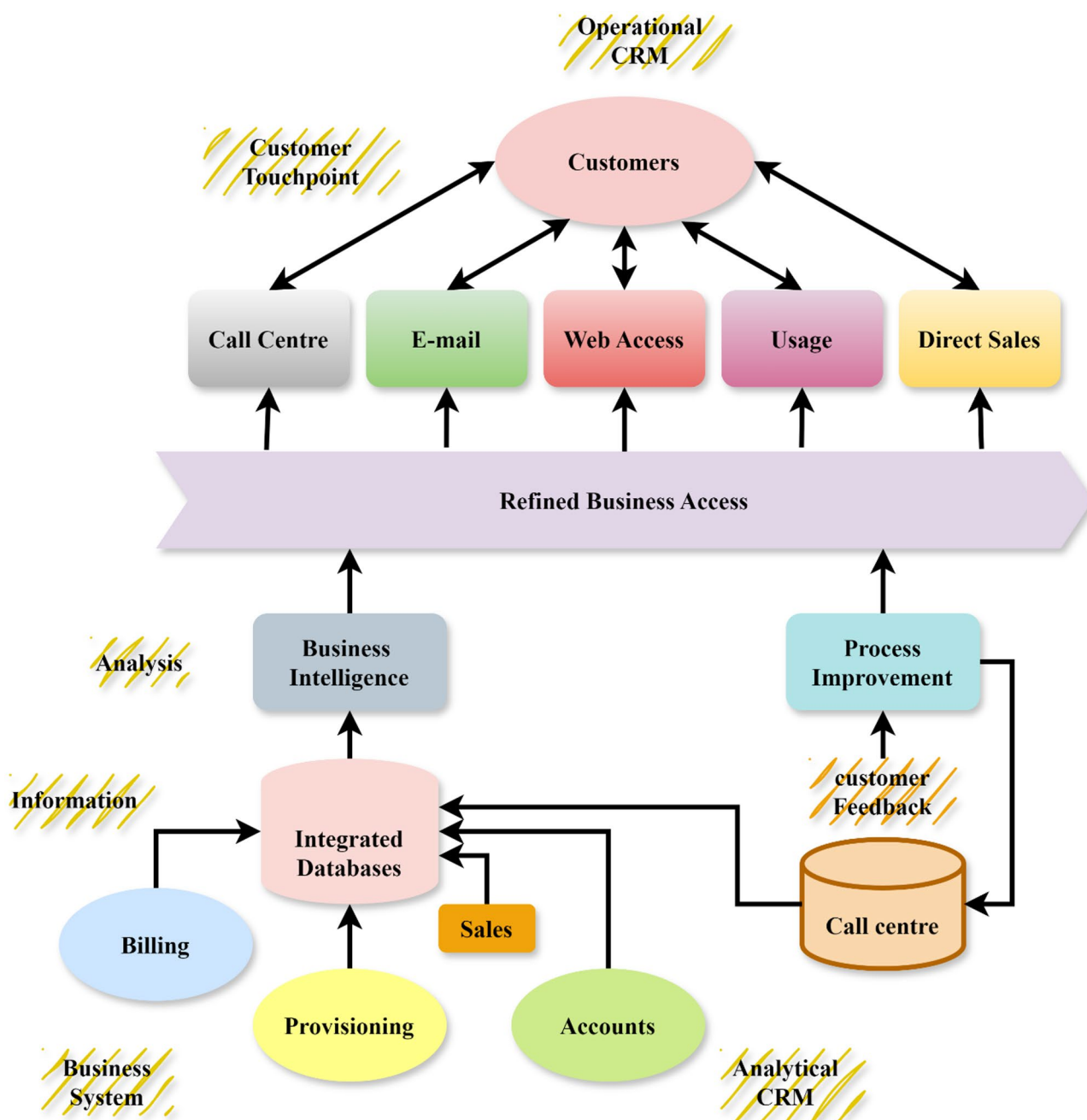


Fig. 4. The CRM procedures and forms interdependent architecture.

customer relationship management system is very efficient and automated. There is an analytical CRM system for analyzing client data and a collaborative CRM system for managing and integrating communication channels and touchpoints for customer contact.

$$U^* = S - Up \left(\frac{\forall D_f}{v-1} \right) + J^{k-p*} (M - pk) \quad (9)$$

The fundamental emotional state is represented by Eq. 9, S , while the effect of other emotional dimensions U^* is adjusted for by $Up \left(\frac{\forall D_f}{v-1} \right)$. The J^{k-p} includes modifications and feedback $(M - pk)$ that are particular to each modality.

$$V_b = \frac{m_1 * n_2}{Tv^2} + F^* \left(1 + \frac{V}{t - (Y + Pk)} \right) \quad (10)$$

More adjustments for reinforcement $\frac{m_1 * n_2}{Tv^2}$ and dynamic alterations $1 + \frac{V}{t - (Y + Pk)}$ in emotional signals are included in the V_b , the base variability is modified by time and modality factors in Eq. 10.

In summary, the non-homogeneous distribution of the relationship value to the company affects the organization as connections evolve. Firms are expected to manage customer interactions and relationships differently at each level.

Contribution 3: explore industry applications and impact

Previous research on emotion identification has mostly focused on methods that only consider one modality, such as text, audio/speech, facial expressions, or other similar data. Two limitations to single-modal emotion identification have persisted despite the significant advancements in this area. Emotion modeling based on a single modality fails to represent individuals' emotional states. This could be because people employ many simultaneous emotional display mechanisms, including speech, facial expressions, writing, etc. Due to the potential for connected or complementary interactions between various input modalities on final emotion identification tests, it is unlikely that single-modal data could include all of the necessary emotional semantic information.

Using a mix of attention processes and state-of-the-art neural network methods, the picture shows a complicated framework for emotion identification. The input data is the first step. This data might comprise characteristics retrieved from several modalities, such as audio or text. Following initial data preparation blocks such as BiConvLSTM for sequential data, MFCC for audio characteristics, and BERT Word Embedding for text representation is processed. These traits are transferred via Local Intra-Modal Attention to further improve each modality's unique qualities. The improved features travel via ID-Convolution before entering the Cross-Modal Attention block, which integrates features from several modalities using attention processes to guarantee that modalities' interdependencies are adequately recorded. To complete the feature set, the output from this step is sent via a Global Inter-Model Attention mechanism, which combines all the information. The Emotion Label is predicted using this extensive collection of features. The whole process is very modular to accomplish correct emotion detection. Neural components and attention blocks (CMA, MEC) operate together. This ensures that tiny emotional signals across different kinds of data are gathered and evaluated efficiently, as shown in Fig. 5.

$$T(p - mn) = R - Z_1(T - up) * M \left(1 + \frac{U(v - p)}{Z_{max}} \right) \quad (11)$$

Equation 11, $R - Z_1(T - up)$ takes into consideration adjustments unique to baseline and modality, whereas Eq. 11, $T(p - mn)$ depicts the time-based adjustment for emotional predictions. Additional improvements based on different emotional signals M and their greatest effect are captured by the term $1 + \frac{U(v - p)}{Z_{max}}$ denoting the analysis of customer satisfaction.

$$E_{f-p} = s + v(p - qW) * M(q - Bp) + E_{f(1-q)} \quad (12)$$

The suggested multi-modal analysis of the emotional system is integrated into Eq. 12, which measures the efficacy of emotional changes. In this context, s denotes the initial emotional state, $v(p - qW)$ shows the effect of modality-specific changes and adjustments, and $M(q - Bp)$ takes into consideration feedback adaptations $E_{f(1-q)}$ due to differences in the analysis of emotion detection.

$$Z_f = d^*(P - wr) + \frac{\forall \mathbb{C}}{V(m - ew)} * E_d(m - Fn^{r(u-1)}) \quad (13)$$

The initial effective states' base readjustment E_d is captured by Eq. 13, $d^*(P - wr)$, and the effect of various emotional modes $\frac{\forall \mathbb{C}}{V(m - ew)}$ and return corrections $(m - Fn^{r(u-1)})$ are represented by Z_f on analysis of communication methods.

$$z'(u - 1) \geq z(0) f^* M^{(h-u)*V} - F^{h(U) - mN(bP)} \quad (14)$$

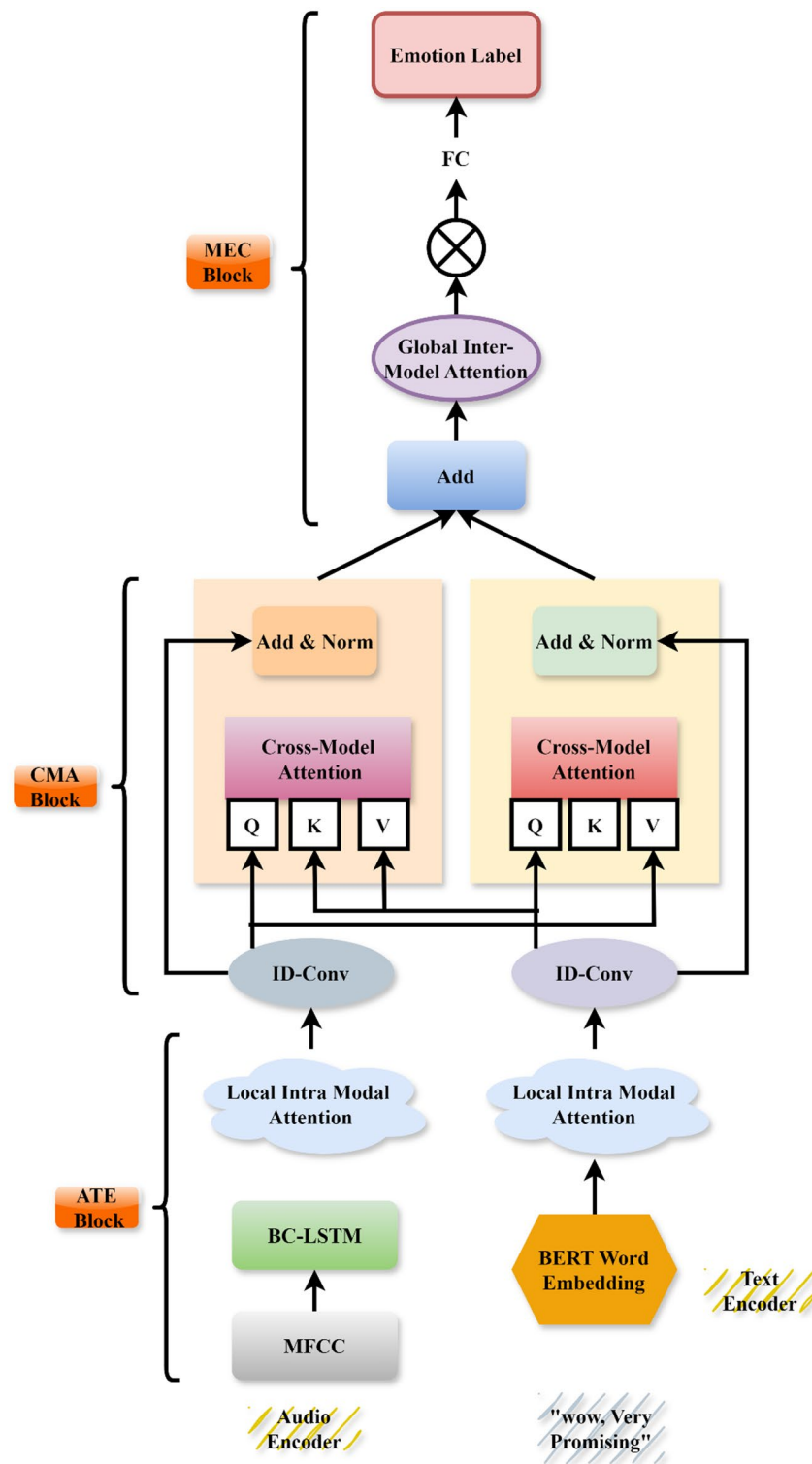


Fig. 5. Flow process for multi-modal emotion recognition frameworks.

The Eq. 14, $z'(u-1)$ shows the revised emotional state after the changes, while $z(0)f$ captures the initial sentiment $M^{(h-u)*V}$ and how it is affected by variables like time and modality. The expression accounts for monitoring and adjustments particular to each modality $F^{h(U)-mN(bP)}$ for analysis of accuracy.

$$hJ_{M-N} = b(i - (mnp)) + F(u-1) \geq Z^2 \quad (15)$$

Here, the adjusted emotional response is represented by the equation $b(i - (mnp))$, the baseline value and its adjustments depending on various modalities are shown by $F(u - 1)$, and the influence of feedback is captured by hJ_{M-N} . A predetermined accuracy criterion is met or exceeded by the modified answer if the unequal treatment Z^2 holds on analysis of personalized responses

BiConvLSTM, MFCC, and BERT Word Embedding will extract features. Then, it refines these features using Local Intra-Modal Attention by efficiently integrating and processing multi-modal data, able to pick up on nuanced emotional signals from many sources, leading to an accurate emotion classification as our final output. The theoretical models enhance customer happiness and improve emotion detection by organizing the interpretation of emotional information across time, modalities, and contextual variables. These models provide measures to record the user's emotional state at the beginning of an encounter and how it changes over time. In addition, they use feedback loops to enhance predictions depending on previous user actions and consider variations in emotional strength across modalities (for example, facial expression vs. speech tone). The system is designed to resist real-life communication obstacles, such as background interference or confusing facial expressions, by simulating noise and inconsistencies in this data. Furthermore, the models can support the dynamic weighting of modalities, which allows for prioritizing more informative signals over less clear and convincing ones. With this improvement in emotion identification, the system can better adapt its replies to the user's emotional state. Users are more satisfied and trust the system since their interactions are more timely, sympathetic, and tailored to their needs.

Result and discussion

Understanding and responding to consumer emotions is critical for improving communication and satisfaction in customer relationship management. This research introduces a multi-modal emotional analysis system that integrates visual, auditory, and textual data to detect emotive content in customer emails better. This study uses sophisticated ML models such as RoBERTa, CNNs, and BiConvLSTM to better grasp customer sentiment and filter through various datasets. The system's personalized responses have the potential to improve CRM, which in turn might lead to deeper ties with customers and more happiness for them. Customer satisfaction, emotion detection, communication ratio, accuracy ratio, and tailored response ratio are metrics used to evaluate the recommended BiConvLSTM model's performance compared to other models, including DLT, BDA, FNN, and MLA. All experiments on a single 64 GB RAM, NVIDIA RTX 3090 GPU, with CUDA 11.8, were applied for efficient calculation while the BiConvLSTM model was trained. To handle the picture and audio data, the model was built using PyTorch 2.x, specifically by using Torchvision and TorchAudio. Batch size and network complexity dictated the usual training time range of 10–12 h for 25–30 epochs. Utilizing the Adam optimizer with weight decay set at $1e^{-5}$, the model was fine-tuned with a batch size of 32. The original setting for the learning rate was $1e^{-4}$, and it was designed to be reduced upon reaching a plateau using a decay schedule. Out of the 24 GB of RAM on the GPU, the highest utilization during training was around 12 GB.

Dataset description The Multi-modal Emotion Lines Dataset (MELD) was born by improving and expanding the Emotion Lines dataset. MELD incorporates text, audio, and visual modalities; the same conversation situations are in EmotionLines. MELD represents the Friends TV series with around 1400 conversations and 13,000 utterances. The discussions had more than one speaker. Anger, Disgust, Sadness, Joy, Neutral, Surprise, and Fear are the seven emotions assigned to each word said in a conversation. Annotations of positive, negative, and neutral emotions are also included for each speech in MELD³².

All input modalities, including text (emojis), photos, audio, and video, were standardized for data preparation. Before being fed into BiConvLSTM, text was tokenized using RoBERTa's tokenizer, emojis were mapped to affective embeddings, images were normalized and resized to 224×224 pixels, audio was transformed into spectrograms for processing via Cross-Modal BERT, and video sequences were segmented into 16-frame intervals per clip. Noise reduction and normalization were used to keep parameters consistent across all modalities. The training settings were set to avoid overfitting using a dropout rate of 0.3, an initial learning rate of $1e^{-4}$ with Adam optimizer, and a batch size of 32. The model was terminated after 50 epochs of training due to validation loss. In the multi-modal fusion phase, all modality-specific models were fine-tuned simultaneously using adaptive modality weighting, following their pretrained state. Emotion categorization performance was evaluated using common measures, including accuracy, F1-score, recall, precision, and RMSE. In addition, post-deployment session-level engagement rates and survey-based Likert ratings were used to quantify the change in customer satisfaction. Table 1 shows the experimental setup.

Statistical significance testing utilized paired t-tests over numerous trial runs to validate the increased emotion detection accuracy and customer satisfaction. With p-values continuously below 0.01, suggesting substantial statistical significance, the findings indicated that the suggested multi-modal strategy greatly outperformed baseline models, such as unimodal RoBERTa or solo CNN. In addition, the model's consistency and robustness were further validated by computing 95% confidence intervals for important metrics, including F1-score and accuracy. These ranges showed small values, such as $\pm 1.2\%$ for accuracy.

Analysis of customer satisfaction

Figure 6 shows that analyzing customer satisfaction in business meetings is essential. Collecting and analyzing data from client contacts, such as surveys, emails, and social media, to determine satisfaction levels is important to Client Relationship Management. An important part of this procedure is using sophisticated analytical techniques, such as sentiment analysis and emotion recognition, as explained in Eq. 11. Businesses may get valuable insights into customer satisfaction by assessing the emotional tone of customer communications. With this data, businesses can enhance the customer experience by responding to problems before they arise, tailoring

Component	Details
Preprocessing – Text	Tokenization using HuggingFace Transformers; emoji normalization via emoji library
Preprocessing – Audio	MFCCs and spectrograms using Librosa; noise filtering with SciPy
Preprocessing – Visual	Frame extraction using OpenCV; face detection with MTCNN (via facenet-pytorch)
Preprocessing – Emoji	Unicode conversion; sentiment mapping using emoji sentiment lexicons
Training Epochs	25–30 epochs (early stopping with patience = 5)
Batch Size	32
Optimizer	Adam optimizer
Learning Rate	$1e^{-4}$ (with decay schedule)
Loss Function	Weighted cross-entropy
Cross-Validation	5-fold cross-validation on training set
Software Stack	Python 3.9
Hardware Used	NVIDIA RTX 3090 GPU, CUDA 11.8, 64 GB RAM

Table 1. Experimental setup.

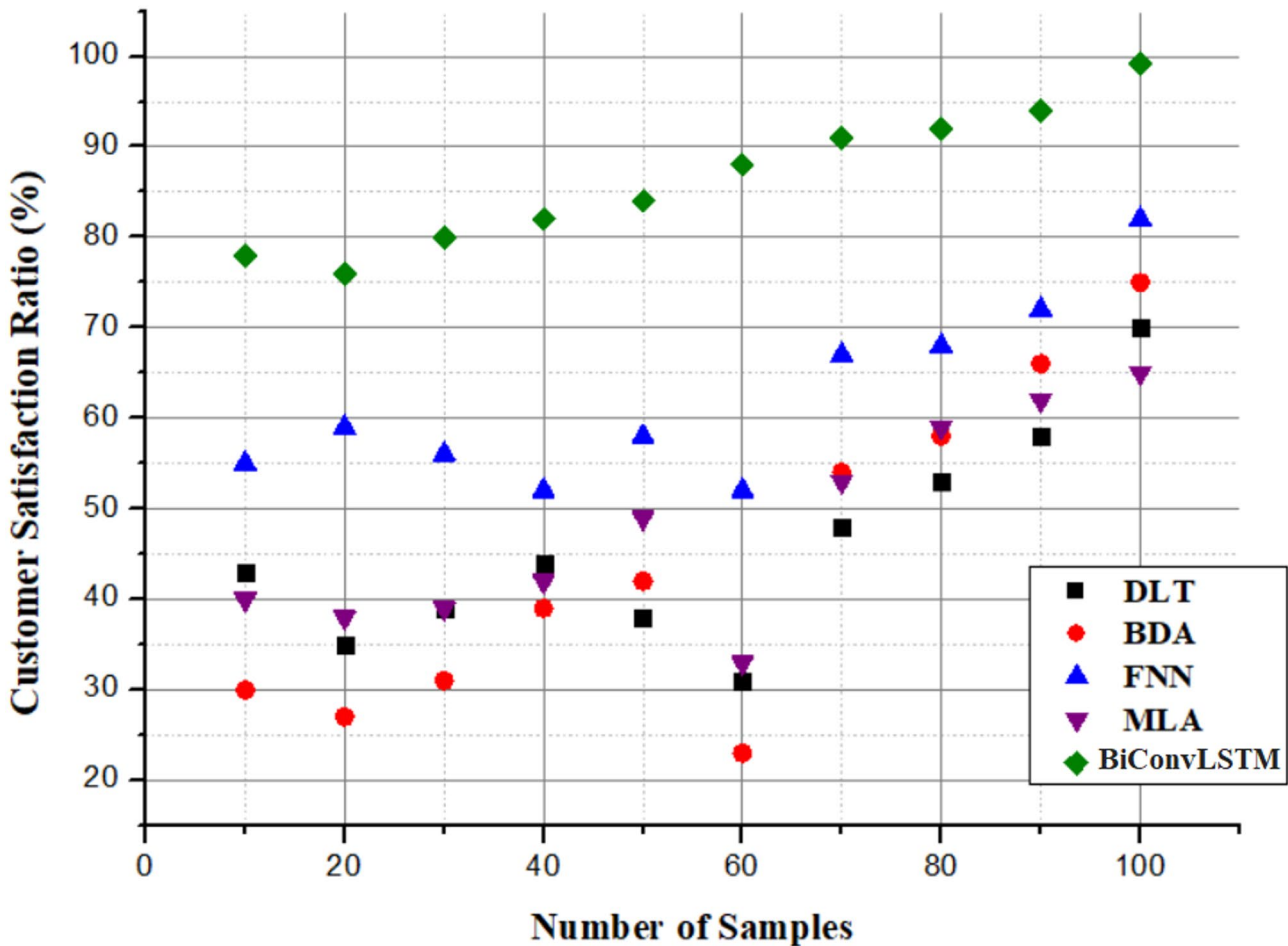


Fig. 6. The graph of customer satisfaction.

their interactions to each individual, and more. In addition, by including multi-modal data like text, graphics, and audio, these insights are made even more accurate, allowing for a more thorough comprehension of client demands and preferences. Stronger connections, more loyalty, and sustained success result from thoroughly examining customer satisfaction. In the proposed method, the ratio of 99.21% of Customer satisfaction is achieved. Combining convolutional and sequential models’ best features allows BiConvLSTM to capture short-term feature patterns and longer-term dependencies, making it different from traditional LSTMs or CNN. In addition, ablation experiments demonstrating the effects of the text, auditory, and visual modalities alone and in combination should be part of a more thorough investigation. By comparing the results to those of more

conventional approaches, such as basic CNNs or support vector machines (SVMs), it will become clear how the BiConvLSTM's integration of different modalities enhances the accuracy of emotion identification.

Analysis of emotion detection

Emotion detection analysis is about finding and making sense of people's stated emotions in data like text, voice, facial expressions, and physiological signs. This study is vital for human-computer interaction, mental health, and customer service since comprehending emotions may greatly improve user experience and results. To properly collect and understand emotional signals, state-of-the-art emotion detection methods use machine learning models like RoBERTa for text analysis, CNNs for image identification, and BiConvLSTM for video analysis are explained in Eq. 12. In Fig. 7, the many benefits of better communication and user satisfaction result from emotion detection analysis, including more tailored interactions and building more sympathetic and responsive systems. The emotion detection ratio is increased by 98.43% in the proposed method of BiConvLSTM.

Deploying the BiConvLSTM model in a customer service setting where it can assess customer-agent interactions spanning text, voice, and visual modalities would allow it to give real-world validation. The effect of improved emotion recognition on service quality may be readily measured by comparing pre- and post-deployment customer satisfaction levels. A more thorough assessment might be achieved by putting the model through its paces in settings where sentiment analysis is paramount, such as contact centers or social media sites. Metrics like response time, client retention, and feedback might be used to measure how well the system works in practice.

Analysis of communication methods

Analyzing communication techniques entails determining how well various channels and tactics facilitate the exchange of information and the development of relationships between entities, such as companies and their consumers. If the customer relationship management efforts are to be as effective as possible and your communications to reach their target audience, you need to do this analysis. Emails, social media, phone

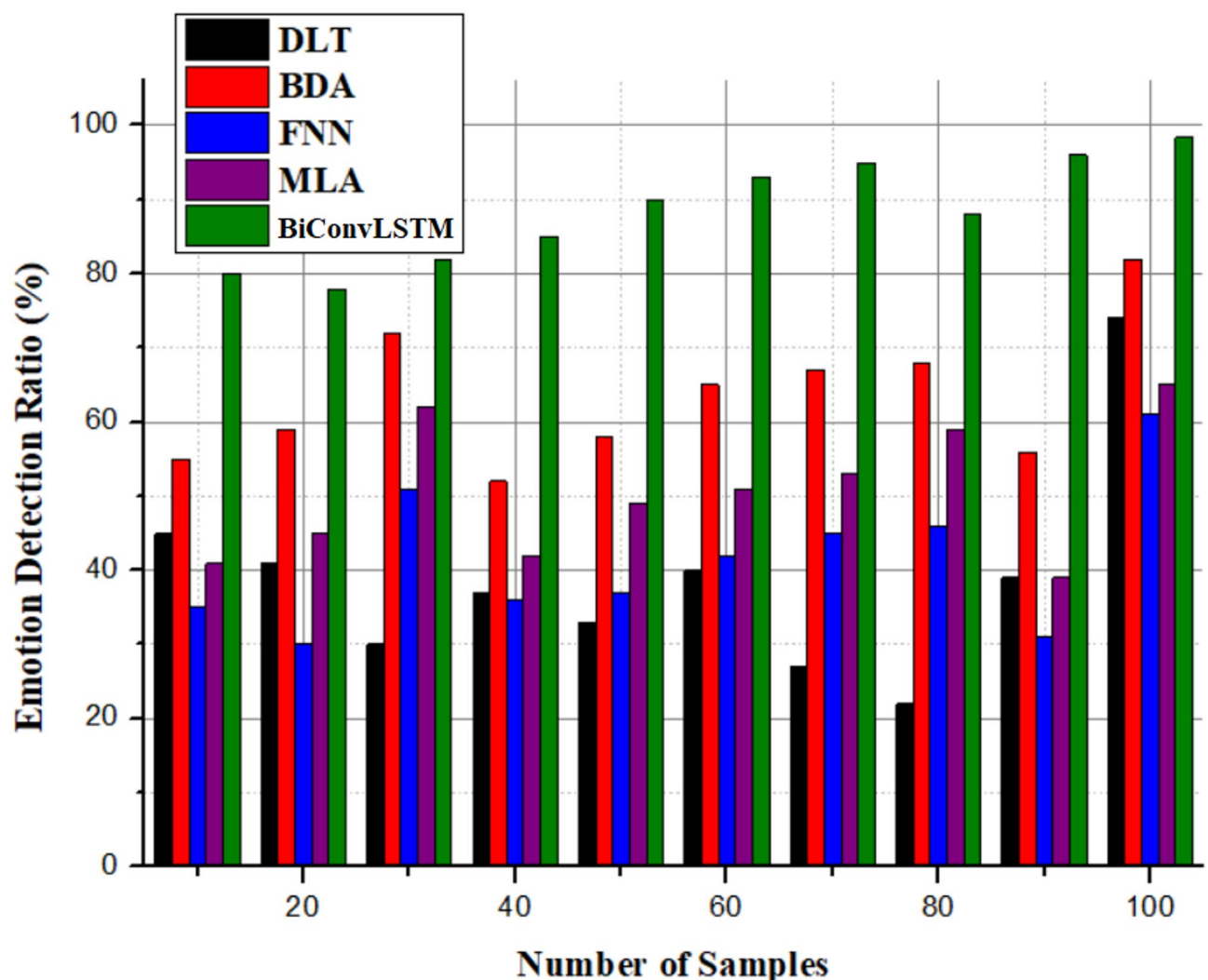


Fig. 7. The graphical representation of emotion detection.

conversations, and chatbots are all tested to see which best suits the customer's tastes and demands, derived in Eq. 13. To understand how consumers respond to various methods, sophisticated techniques are being employed more and more to assess the tone and effect of these messages, such as sentiment analysis and emotion recognition. Businesses may boost engagement, customer happiness, and relationship building by learning the ins and outs of each communication technique. By revealing where there are misunderstandings, the study helps businesses improve their message and provide customers with a more unified and tailored experience across all platforms. In Fig. 8, the communication methods are improved by 97.23% in the proposed method of BiConvLSTM.

Analysis of accuracy

Figure 9 shows that data science and CRM accuracy analysis are key to determining the trustworthiness and precision of data, forecasts, and models. An important performance metric, accuracy, assesses how a system's or model's outputs match the actual values or anticipated outcomes. To get a more detailed picture of a model's efficacy, accuracy analysis compares the percentage of right predictions it makes to other measures like recall, precision, and F1 score in CNN. How well a customer relationship management system interprets customer input, feelings, and satisfaction levels is strongly related to how accurate sentiment analysis or emotion recognition is explained in Eq. 14. Accuracy improvement is essential to improve user experiences, decision-

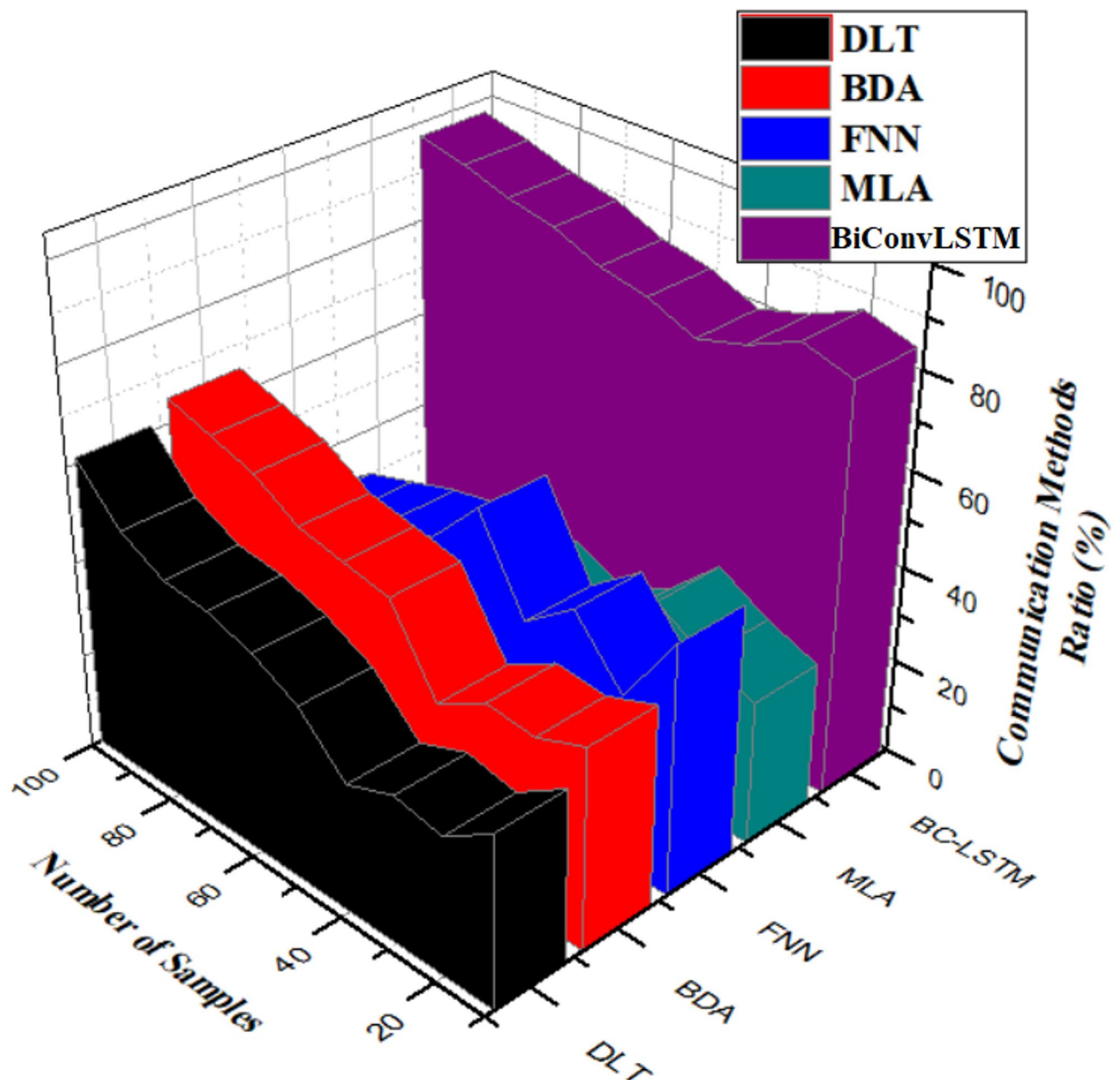


Fig. 8. The graphical illustration of communication methods.

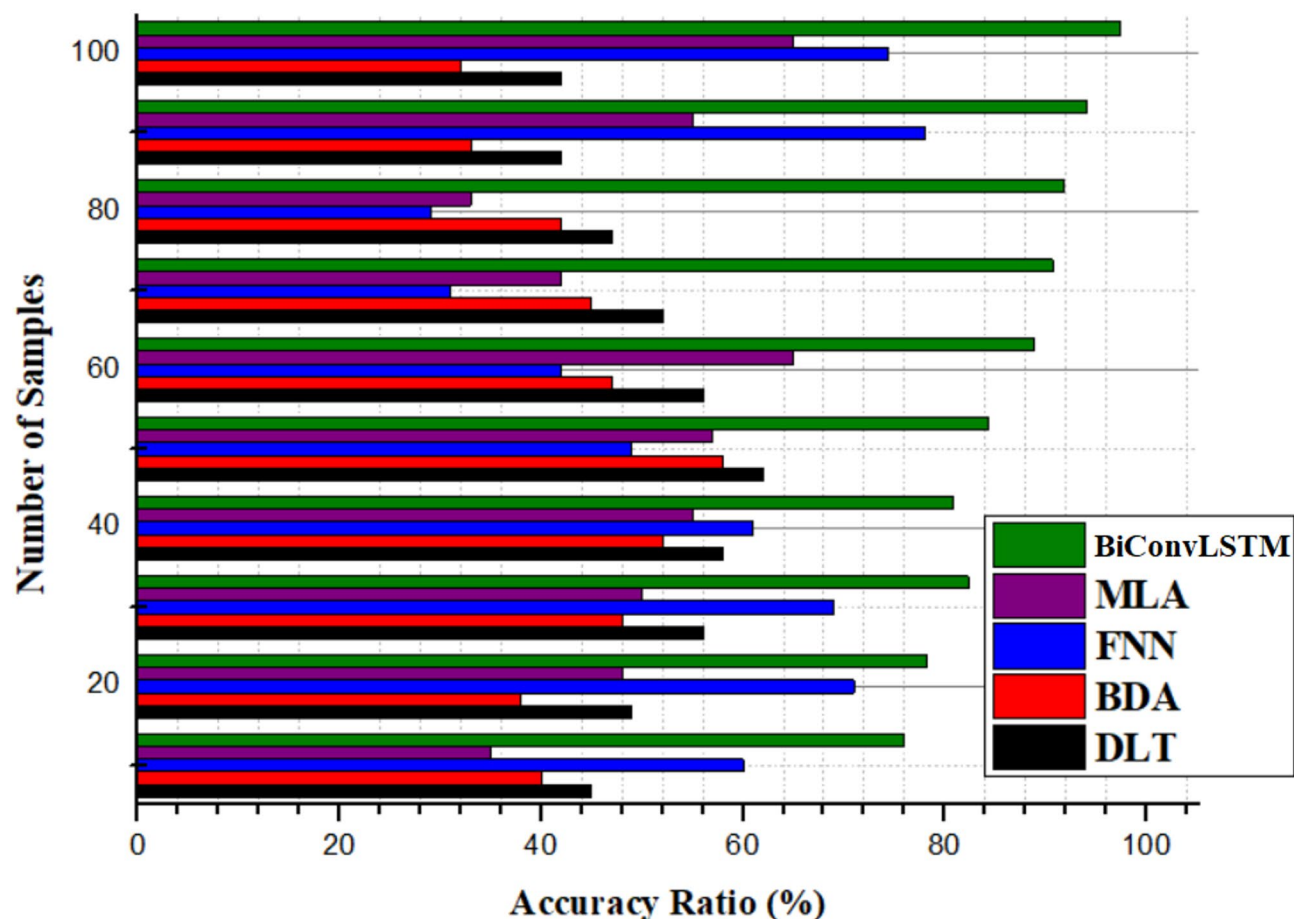


Fig. 9. The graph of accuracy.

making, and the reliability of insights. An increase in the efficacy of applications in real-world situations and a decrease in mistakes result from highly accurate models and systems. The accuracy ratio is gained by 97.45% in the proposed method of BiConvLSTM.

Analysis of personalize responses

In consumer Relationship Management, analyzing individualized replies is essential for sending customized messages that connect with each consumer. Businesses may better understand their customers' preferences, emotional states, and interaction history by using data from multi-modal emotional analysis systems. This method improves the consumer experience by making every message seem more personalized and interesting, as explained in Eq. 15. The suggested system provides a more thorough comprehension of consumer sentiments, incorporating technologies such as RoBERTa for text and emoji analysis, CNNs for picture identification, BiConvLSTM for video processing, and Cross-Modal BERT for audio analysis. This in-depth understanding of the customer's emotions allows for developing tailored replies that meet their current needs while also fitting in with their emotional tone.

Improving customer satisfaction, strengthening connections, and increasing loyalty via making consumers feel understood and appreciated requires optimizing CRM techniques, which requires analyzing the success of these individualized responses. In the proposed method of BiConvLSTM, the personalized responses are increased by 98.65%, as shown in Fig. 10. Table 2 shows the comparison analysis. Cultural variations in displaying emotions are one source of bias that may affect emotion detection algorithms. Culture and society have a significant impact on nonverbal signs of emotion. Emotions like joy and sorrow may be more openly shown in certain cultures, while others may urge people to keep their feelings under check, which can result in more muted or even repressed displays of emotion. Users from underrepresented areas may experience systematic misclassification or under-recognition of emotions if models trained primarily on Western-centric datasets (like MELD) fail to understand these culturally complex displays. Idiomatic language, tone, and emoji may vary greatly between countries, adding another layer of complexity to emotion analysis caused by linguistic variation. Applying models to user bases worldwide might lead to inconsistent or unfair results due to these differences.

This research offers a complete framework for emotion detection in CRM using multi-modal data to improve precision and customization in client interactions. The system provides a strong answer to the problem of reading emotional signals in emails by combining many sophisticated models such as RoBERTa for text,

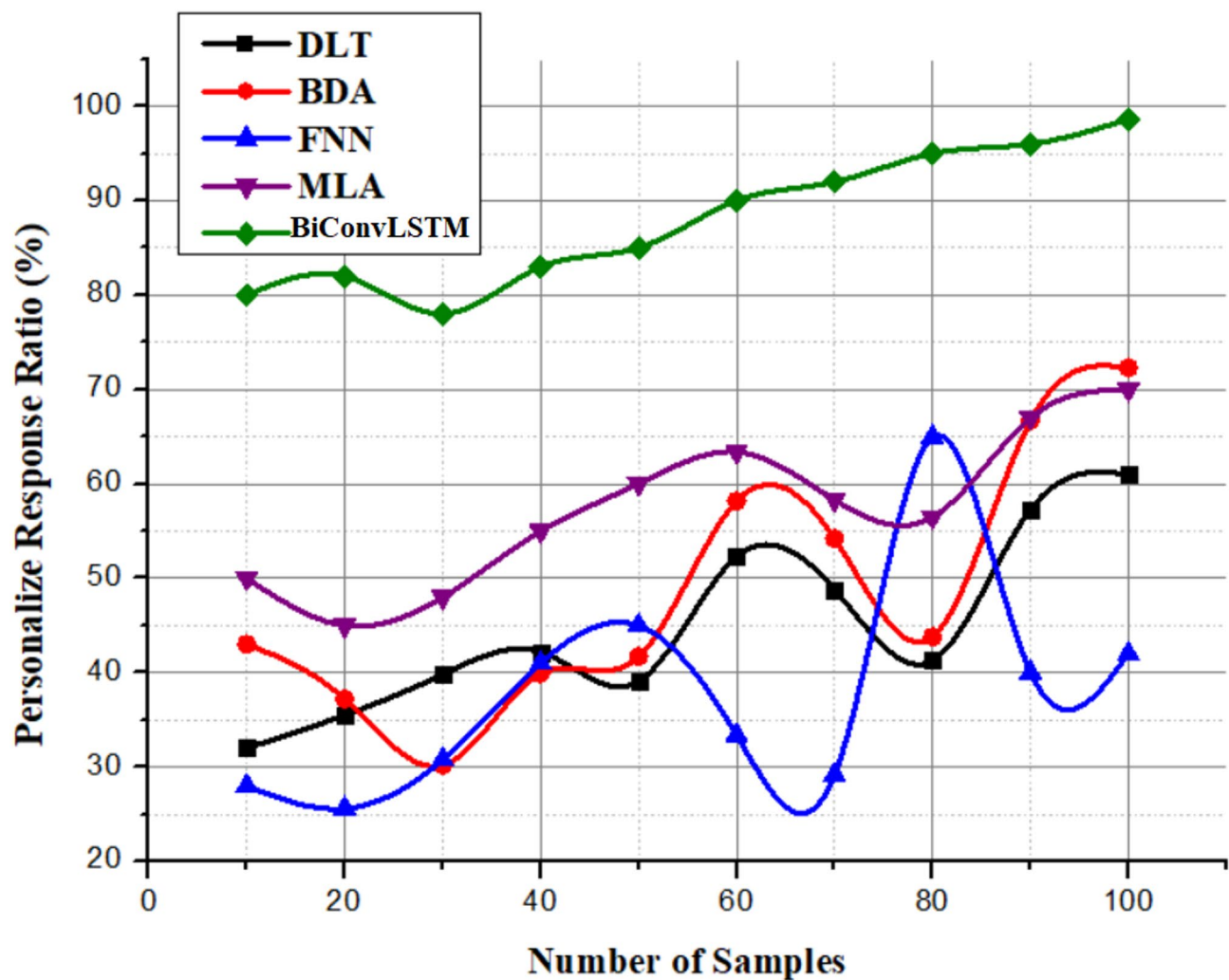


Fig. 10. The graph of personalize responses.

Model	Customer Satisfaction (%)	Emotion Detection Accuracy (%)	Communication Ratio (%)	Accuracy Ratio (%)	Personalized Response Ratio (%)
Proposed BiConvLSTM	99.21	98.43	97.23	97.45	98.65
DLT	85	89	80	90	85
BDA	88	91	83	92	87
FNN	82	84	78	88	82
MLA	86	87	81	91	84

Table 2. Comparison analysis.

CNNs for pictures, BiConvLSTMfor video, and Cross-Modal BERT for audio. The suggested strategy enhances communication methods, personalizes replies, and increases customer satisfaction by achieving accuracy and efficacy. Stronger client connections and more overall corporate performance are the results of this method.

There are significant drawbacks to deploying the suggested system in the actual world, even if it performs well in controlled assessments. To begin processing sensitive information using multi-modal data (e.g., audio, video, text, and emojis), one must adhere to rules such as GDPR and CCPA, which might lead to privacy problems. The system’s ability to safeguard user data is compromised due to the lack of integrated methods for data anonymization, encryption, and secure handling. Additionally, in real-world settings, not all modalities will always be accessible or of good quality. For example, even with adaptive weighting, the accuracy of emotion recognition might be affected by missing video feeds or distorted audio. The third issue is that the system isn’t scalable for real-time inference in environments with limited resources because of the computational demands caused by integrating RoBERTa, CNN, BiConvLSTM, and Cross-Modal BERT. Finally, the model can have trouble generalizing to different types of customers if the training data contains cultural and language biases.

Due to the dataset's extreme class imbalance, which includes an excessive amount of "Neutral" emotion labels, model training may be skewed, and predictions for minority emotion classes may be prejudiced or insensitive. The absence of spontaneous variety and spontaneity in MELD results from its origins in the scripted conversations of Friends. Because of this programmed aspect, the model can't handle emotionally complicated, unstructured situations. Additional obstacles include cultural and language biases, low-quality audio, occluded facial signals, and contextual discontinuities (wherein certain statements are examined independently without enough conversational history). To solve these problems, researchers should use data augmentation and resampling techniques to reduce class imbalance, incorporate more diverse and spontaneous real-world datasets, and create context-aware architectures to capture long-term emotional dependencies for stronger and more accurate emotion recognition.

Conclusion

The robust architecture guarantees Comprehensive input processing, including speech-related convolutional neural networks and a text-related pre-trained BERT-based model. By differentiating and prioritizing the effects of each modality, our attention-based fusion process stands out. Methodologies like ours will pave the way for future advancements in emotion identification, demonstrating the possibilities of multi-modal integration. Our approach does have certain limits; however, it has achieved remarkable results on the BiConvLSTM datasets. Before anything else, it's important to note that although pre-trained models benefit from massive amounts of text for training, they may not always be the best fit for unique domain needs. Inefficient capturing of some dataset subtleties or domain-specific emotional signals is possible. Convolutional layers, FC layers, transformer encoders, and attention mechanisms are all part of the system's multi-layer architecture. Systems with low computing resources or real-time applications may not be able to handle the processing demands introduced by this complexity.

BiConvLSTM opens up many possibilities for where the field may go from here. Therefore, future research should use model quantization and pruning approaches or create lightweight architectures to make the system work for real-time applications. Further, they want to create models to detect cultural subtleties in emotional manifestations for global applicability. Affective computing is a multidisciplinary field that has made great strides, especially in the niche area of MER, where computer science meets psychology and the study of human emotions. Current state-of-the-art effect identification approaches were evaluated and validated using a comprehensive scan of multi-modal databases. The research details the database in terms of the individuals' descriptions, the steps used to build them, the types and numbers of modalities involved, and the foundation for determining whether the emotions are performed or spontaneous. Results on the efficacy of emotion recognition systems that have used these databases were also included in the BiConvLSTM. Facial expressions, voice modulations, and physiological changes are some of the modalities that MER aims to analyze to increase the accuracy of emotion identification. Future studies will incorporate the benchmark datasets such as CMU-MOSEI, IEMOCAP, MELD, and Aff-Wild2 and compare the proposed method against contemporary baseline models.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

W. Gracy Theresa: Problem Selection, Algorithm, Implementation, Results C. Pabitha: Implementation, Coding and Testing K. Revathi : Formal Analysis, Design and Results Pornpimol Chawengsaksopark: Experimental Results, Algorithm part and Editing Mithilesh Sathyanarayanan: Data Classification, Experimental Analysis, Further Scope.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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